Decision Effects and Failure Mechanisms of Post-Selection Error Monitoring in Surrogate-Assisted Black-Box Optimization

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A mechanism study and negative-result analysis under a prespecified gate

Abstract

We examine whether one-sided monitoring of evaluated candidates improves objective-evaluation allocation in surrogate-assisted black-box optimization. A candidate-independent additive correction preserves predicted-mean ordering in exact real arithmetic under the specified support and selection rules; floating-point outcomes are audited separately. The main temporal prototype can therefore change decisions through feedback exploration rather than a new ranking. We isolate common-history calibration, common-state immediate utility, and independent closed-loop optimization, with every diagnostic label excluded from optimizer inputs.

The prototype fails its development optimization-benefit gate over eight BBOB functions and dimensions 10/20. We then freeze an unchanged mechanism validation over all 24 functions, dimensions 10/20/40, instances 11–13, and three repeats. It completes 3,888 internal runs, 907,200 paid evaluations, and 1,990,656 additional diagnostic labels. Relative to the development-selected fixed .1 exploration control, independent endpoint effects are D=10: ratio 1.468, Holm p=5.36e-06; D=20: ratio 1.258, Holm p=0.00170; D=40: ratio 1.159, Holm p=0.00170. Nine prespecified contrasts use one Holm family; unadjusted hierarchical intervals and every function direction are disclosed. History-policy differences, immediate gains, improvement-declaration denominators, localized-bound support, and final error are reported separately. Sobol/CMA-ES provide budget-matched descriptive background evidence. The original failed gate remains unchanged. Conclusions concern this frozen decision rule, model, candidate process, and budget, without a coverage theorem or general high-dimensional claim. Complete GP/background references, exclusive costs, and three-task application evidence are reported separately. The predefined 300-dimensional extension is reported in Section 11. The absence of verifiable implementations for the named competitive SAEAs limits competitive claims rather than leaving the mechanism and negative-result study experimentally incomplete.

Eight methods complete 480 application runs with 500 evaluations each on fixed data/splits; the application-only comparisons yield synt_simple: median raw loss difference +0.0503, Holm p=0.0826; synt_medium: median raw loss difference +0.0393, Holm p=0.0192; DNA: median raw loss difference +0.00271, Holm p=0.0592. A separate EngiBench finite-element beam study completes 50 runs and 10,000 evaluations. PEM-T has median compliance 1972.8 versus 1435.6 for fixed .1 exploration and is worse in 8/10 paired repeats; TuRBO-1 provides the strongest descriptive reference at 578.2. Opti-

mization randomness, independent endpoint re-evaluation, and cross-task scale differences are reported separately.

Keywords: expensive black-box optimization; surrogate-assisted evolutionary search; prequential residuals; post-selection error monitoring; model management

1 Introduction

An objective evaluation may require a simulation, model training, or a physical experiment. In these settings, a surrogate can help select which candidate merits the next evaluation. The surrogate's usefulness depends on how its predictions affect these decisions, rather than solely on its accuracy over an arbitrary test population. A predictor may be accurate over most proposed candidates while performing differently on candidates favored by its own ranking. Equally, an upper bound may rarely fail while rarely identifying useful opportunities.

We consider deterministic, single-objective optimization over a bounded continuous domain. The objective does not drift within a run. The surrogate, its training population, and the candidate distribution nevertheless change as evaluated points accumulate. Historical residuals can thus differ in both their model of origin and their prediction-time scale. Recording them honestly supports an online empirical monitor; it does not by itself establish exchangeability or a conformal coverage guarantee.

Our original question is whether monitoring the error of the actually evaluated candidates can improve objective-evaluation allocation relative to fixed selection rules. The development protocol permits a transition to mechanism analysis if the feedback prototype does not improve optimization. We follow that branch rather than reinterpret a coverage statistic as evidence of optimization efficiency.

The study separates three experiments. Common-history diagnostics compare residual calibrators at identical frozen models and pools. Common-state experiments compare the immediate utility of alternative decisions without updating the history. Independent closed-loop trajectories evaluate the resulting optimization behavior. Extra labels used by the first two experiments are acquired only on inexpensive analytic benchmarks, stored separately, and never provided to the optimizer.

The contributions retained at this stage are narrowly scoped: an ordering-equivalence result for the specified additive correction and selection rule; an auditable implementation of prediction-before-label residual monitoring; and empirical evidence distinguishing calibration diagnostics, improvement declarations, and optimization utility. We do not claim a new conformal construction, superiority to published methods, or general high-dimensional effectiveness.

2 Related work and scope of comparison

Online calibration has already been used in Bayesian optimization. Deshpande, Marx, and Kuleshov [1] study calibrated predictive distributions and online updating. Their approach differs from taking a fixed-weight empirical quantile of historical signed residuals. Its assumptions and guarantees cannot be imported into the present implementation.

LOCBO [2] uses localized online conformal thresholds to calibrate a likelihood, obtains a posterior through denoising, and uses that posterior in optimization. Its guarantees distinguish noisy observations from the latent objective and depend on the construction and stated assumptions. An empirical spatial

residual weight and an effective-sample-size threshold are not equivalent substitutes. Moreover, a constant correction to a mean ranking is not a theorem about all distribution-calibrated acquisition functions.

Temporal conformal prediction has also been studied in surrogate-assisted evolutionary optimization [3]. The publisher's abstract reports temporal weighting and optimization evaluation savings. We have verified that abstract, but have not obtained a reliable full implementation. Consequently, the temporal residual monitor here is an internal reference, not a reproduction of that work. The algebraic redundancy identified below concerns our decision rule and must not be attributed to the published method without its algorithm.

JOMI [4] addresses selection-conditional prediction through a reference-set construction. Its theorem requires a selection rule invariant to permutations of the calibration data and a specified conditional exchangeability condition. Our adaptive, cross-model residual history does not have an established justification for those conditions. We diagnose selected-point errors but do not construct JOMI prediction sets or claim its validity.

SADE-SS [5] studies parameter adaptation, search-space tightening, and global/local search management. Boosting Gaussian-process uncertainty [6] and the use of performance, spatial, and temporal evolutionary knowledge [7] provide further context. Full text or reliable executable details remain incomplete for several of these methods. This manuscript therefore reports internal mechanism comparisons and keeps the competitive evidence gap explicit.

3 Problem, information, and decision events

The problem is

$$\min_{x\in\Omega} f(x), \qquad \Omega = [l,u]^D,$$

with a budget of B objective calls. Initialization, online startup, repeated calls, and subsequent selected evaluations all count. No constraints, observation noise, multiple fidelities, or multiple objectives are included.

At evaluation t, the available labeled history is $\mathcal{D}_{t-1} = \{(x_i, y_i) : i < t\}$. Before obtaining y_t, we persist the selected point, its predicted mean $\mu_t(x_t)$, empirical upper bound $H_t(x_t)$, incumbent b_{t-1}, prediction-time scale s_t, selection mode, and random state. The journal is flushed before the objective is invoked. The signed prequential residual is

$$r_t = \frac{y_t - \mu_t(x_t)}{s_t}.$$

The scale is the population standard deviation of the current training labels, replaced by one when it is zero. Each residual retains the original prediction and scale. We never recalculate it with a later model or use a training residual as if it were an online prediction error.

An upper-bound failure is $e_t = \mathbb{I}[y_t > H_t(x_t)]$. A monitored improvement declaration requires sufficient support and

$$H_t(x_t) < b_{t-1} - \tau_t, \qquad \tau_t = 10^{-6} s_t.$$

It is false if $y_t \geq b_{t-1} - \tau_t$. Every false declaration is an upper-bound failure on that declared point. This elementary event inclusion does not control the declaration denominator, establish a per-declaration success probability, or imply a bound on the false-discovery proportion. A run with no declarations has an undefined FDP, reported as NA.

4 Search framework and empirical monitoring

4.1 Initialization, model, and candidates

We use a randomized Latin hypercube with $n_0 = 2D + 20$ points in a fixed order. The first $D + 10$ points train the initial model. The next $D + 10$ points are evaluated sequentially after prediction, forming online startup residuals. Internal variants share this initialization and startup process.
Inputs are scaled to $[0, 1]^D$. The regularized Gaussian RBF uses

$$d(x, x') = \|x - x'\|_2/\sqrt{D}, \qquad k(x, x') = \exp\{-d(x, x')^2/(2 \cdot 0.25^2)\},$$

and ridge 10^{-6}. Labels are standardized using the current training subset. At most 256 points are retained: up to 128 best-valued points followed by farthest-point sampling from the remaining archive. Below the cap, all observations are used. This is an implementation choice to limit dense linear-algebra costs, not an RBF complexity advantage.

Each round proposes 256 candidates: 128 reflected DE/current-to-pbest/1 proposals with $F = .5$, $CR = .9$, and $p = .2$, and 128 independently scrambled Sobol points. The parent pool has at most 64 members, combining elite and spatially dispersed historical points. Candidates within 10^{-10} of evaluated points or duplicates are removed and replenished from a separate Sobol stream. Ties are resolved by the original index. One candidate is evaluated per round. Random streams are keyed by seed, purpose, and evaluation index; the core search does not use the total budget.

4.2 Residual upper bounds and support

For candidate x, the empirical bound is

$$q_t(x) = \max\{0, Q^w_{.9}(r_i : i < t)\}, \qquad H_t(x) = \mu_t(x) + s_t q_t(x).$$

The quantile is the smallest observed residual at which cumulative normalized weight first reaches .9, without interpolation. We compare four weight rules over the most recent 100 residuals: global equal weights (G), temporal weights with a half-life of 25 evaluations (T), spatial Gaussian weights (S), and their product (ST). Spatial bandwidth is the twentieth-neighbor distance. The effective sample size is

$$n_{\text{eff}}(x) = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}.$$

Below 20 effective samples, ST falls back to T, S to G, and insufficient T to G. Fewer than 20 total residuals disables bound-based improvement selection. Support, fallback, and weighted distances are recorded. This threshold checks weight concentration and is not a representativeness or coverage guarantee.

4.3 Feedback and selection

Over at most the last 30 online predictions, let $\bar{e}_t$ be the failure fraction and $\bar{v}_t$ the mean normalized margin, where $v_i = (H_i - \mu_i)/s_i$. Define

$$a_t = \max\left\{\text{clip}\left(\frac{\bar{e}_t - .1}{.2}, 0, 1\right), \text{clip}\left(\frac{\bar{v}_t - 1}{2}, 0, 1\right)\right\}.$$

Let h_t count evaluations without significant incumbent improvement and $z_t = \min(h_t/20, 1)$. The prototype uses

$$p_t = \text{clip}(.1 + .4a_t + .3z_t, .1, .8).$$

With fewer than 20 monitoring records, $p_t = .5$ and the improvement mode is disabled. Otherwise, spatial exploration with probability p_t selects the candidate maximizing its minimum distance to the evaluated archive. On the complementary branch, if supported improvement candidates exist and $a_t < .5$, the candidate with the smallest supported empirical bound is selected; otherwise the smallest predicted mean is selected. The coefficients are frozen development choices, not theoretically optimal values. Higher error, width, or stagnation can induce more exploration, but whether those evaluations benefit optimization is an empirical question.

5 Structural audit and limitations of event reasoning

Ordering equivalence in exact real arithmetic. If support is common across candidates, $q_t(x) = q_t$, and s_t is a common scalar, then $\arg\min_x H_t(x) = \arg\min_x \mu_t(x)$ under the same tie rule. Adding the common constant $s_t q_t$ preserves every pairwise comparison. Checking whether an improvement candidate exists does not change the minimizer in the studied selection rule. Thus G/T upper-bound and mean-greedy modes select the same point on the non-exploration branch.

If the exploration probability and random streams are also unchanged, removing the upper mode produces the same entire trajectory by induction over evaluations. Temporal monitoring with a fixed probability likewise produces the corresponding fixed mean/exploration trajectory. These configurations are retained as implementation audits, not independent mechanism ablations. Spatial bounds can vary across candidates and need not preserve the mean ranking. Candidate-dependent support, other acquisition functions, evaluation-trigger decisions, or altered stopping policies are outside this equivalence statement.

False declarations. If $H_t < b_{t-1} - \tau_t$ and $y_t \geq b_{t-1} - \tau_t$, then $y_t > H_t$. No distributional argument is required. Conversely, an overall failure fraction of .1 can coexist with a false-declaration fraction of one if all failures fall in a rare declared subset. The supplied deterministic counterexample distinguishes this event relation from a reliability guarantee; it is not included in benchmark statistics.

6 Development protocol and statistical units

We preselected BBOB functions 1, 3, 6, 10, 15, 20, 21, and 24, dimensions 10 and 20, instances 1–3, and three repeats per instance. Each configuration has 144 runs. The budget is $10D$, including initialization. The primary prototype is T; it is not chosen after ranking the spatial variants.

The closed-loop experiment compares 13 configurations: mean greedy, fixed exploration probabilities .1/.3/.5/.8, temporal monitoring with fixed .3 management, PEM-G/T/S/ST feedback, removal of the upper mode, failure-only feedback, and stagnation-only feedback. The failure-only variant removes both width feedback and stagnation, so it cannot identify those two terms' separate causal contributions. Mean-greedy and fixed strategies still compute T bounds for their monitoring journals, but the bounds do not affect their decisions.

Common histories use mean greedy, fixed .3 mixing, and uniform selection from the current 256-candidate pool. Pool-random is not global independent uniform sampling, and maximum-distance ex-

ploration is not a random history. At four frozen evaluation indices we compare G/T/S/ST and alternative single-step choices using independently acquired pool labels. The first checkpoint is moved beyond startup when needed: 41/50/75/100 in 10 dimensions and 61/100/150/200 in 20 dimensions. These corrected first checkpoints are not described as 25% budget.

Let $E_B = \max(f_{\text{best},B} - f^*, 0)$ and let E_{n_0} be the common initial error, floored at 10^{-12}. The endpoint is $\log_{10} \max(E_B/E_{n_0}, 10^{-12})$. For each function/instance, we take the median over three runs, then average the three instances; functions receive equal weight within each dimension. The paired difference is prototype minus control, with negative values favoring the prototype. Its base-10 exponent is a geometric summary of relative errors, not a ratio that every run shares.

The fixed probability is selected only on development data, by equal-weight endpoint scores over dimensions and functions. Eight function-level paired contrasts per dimension enter Wilcoxon tests. Twenty comparisons are jointly Holm-adjusted. Four thousand paired hierarchical bootstrap resamples draw functions, instances, and common run indices; each method's run median is recomputed before subtraction. Bootstrap intervals are not multiplicity-adjusted and do not remove uncertainty introduced by selecting the fixed control on this dataset. No development comparison is represented as independent confirmation.

Diagnostics report pool, predicted top 10% (26 candidates), actual selected point, strategy-selected point, and the declared subset separately. Failure fractions are first averaged by state, while sparse declarations additionally disclose nonempty states, candidate counts, and pooled failures. Candidate labels within a pool are correlated and are not treated as independent statistical problems.

7 Audited development results

7.1 Completion and event audits

All 1,872 development runs complete without exceptions, charging 280,800 objective calls. The 432 common histories charge 64,800 calls. Their four checkpoints yield 1,728 states and 442,368 separate diagnostic labels. The two batches therefore execute 2,304 runs and 345,600 algorithm calls. Greedy and fixed .3 common histories repeat 288 development trajectories to capture full pools; these executions do not increase the optimization statistical sample. Optimization statistics use only the 1,872 development runs. Prediction-before-label ordering, common initialization, and unchanged diagnostic-source journals pass the audit.

PEM-T and the removed-upper-mode variant have identical trajectories in all 144 pairs. Fixed .3 and temporal monitoring with fixed .3 management also match in all 144 pairs. These are correspondence checks for the structural result, not two independently effective components.

7.2 Endpoint behavior and the development gate

The selected fixed probability is .1 among .1/.3/.5/.8. Table 1 gives function-equally-weighted log-relative endpoints; lower is better. The complete feedback prototype and the spatial version do not outperform that fixed control on this summary. Mean greedy is slightly better than fixed .1 in both dimensions.

Table 1: Function-weighted development endpoint scores

D	Greedy	Fixed .1	PEM-T	PEM-ST	Error only	Stagnation
10	-1.3552	-1.3499	-1.0668	-1.0866	-1.2898	-1.2613
20	-1.3324	-1.3211	-1.1401	-1.1097	-1.2851	-1.1757

Table 2 reports PEM-T minus fixed .1. The geometric ratios exceed one and each dimension has zero wins and eight losses. Paired hierarchical bootstrap intervals for the log difference are D=10: [0.060, 0.484]; D=20: [0.012, 0.521]. The Holm tests do not reach .05. The intervals and adjusted tests address different procedures and must not be selectively combined into a confirmatory significance claim; the control is also selected on development data.

Table 2: Paired function-level development effects

D	Log difference	Geometric ratio	Wins	Losses	Holm p
10	0.2831	1.9192	0	8	0.1562
20	0.1810	1.5170	0	8	0.1562

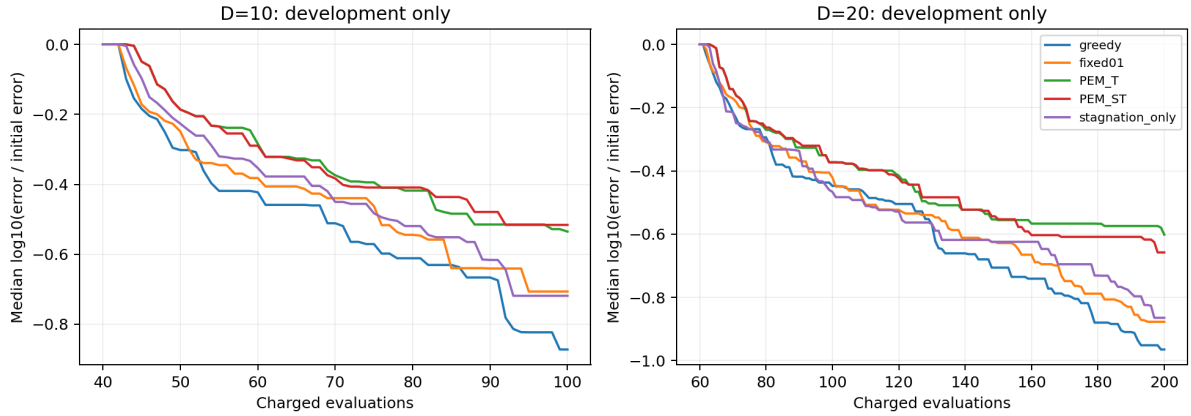


Figure 1: Development convergence curves. Each evaluation pools normalized log errors over functions, instances, and runs using a median. This differs from Table 1's function-level aggregation, so curve endpoints are not Table 1 estimates.

The necessary optimization-benefit gate is unmet in both dimensions. We do not enlarge the original optimization-confirmation experiment to circumvent that result. A later mechanism validation is separately specified, and any future new optimizer would require a new development stage and held-out range.

7.3 Exploration, failure, and sparse declarations

PEM-T explores approximately 36.6% and 33.4% of post-initialization evaluations, compared with approximately 10% for the fixed control. Its observed failure fractions are approximately 9.9% and 9.5%. Being near the nominal monitoring level does not correspond to better endpoints.

Table 3: Joint closed-loop decision indicators, 72 runs per dimension/method

D	Method	Exploration	Failure	Claims	False	Pooled FDP
10	PEM_ST	0.3574	0.1062	7	4	0.5714
10	PEM_T	0.3664	0.0986	3	1	0.3333
10	error_only	0.1574	0.0847	4	2	0.5000
10	fixed01	0.0995	0.0843	4	2	0.5000
10	stagnation_only	0.2338	0.0972	2	0	0.0000
20	PEM_ST	0.3478	0.1098	32	2	0.0625
20	PEM_T	0.3335	0.0946	0	0	NA
20	error_only	0.1475	0.0864	8	4	0.5000
20	fixed01	0.1017	0.0868	10	9	0.9000
20	stagnation_only	0.2394	0.0951	1	1	1.0000

PEM-T makes only three declarations in dimension 10, one false, and no declarations in dimension 20. The latter FDP is NA, not zero. PEM-ST makes 32 declarations with two false in dimension 20, but its endpoint remains worse. Table 3's pooled declaration ratio is distinct from a mean over runs with nonzero declarations. These sparse denominators do not establish stable conditional success probabilities.

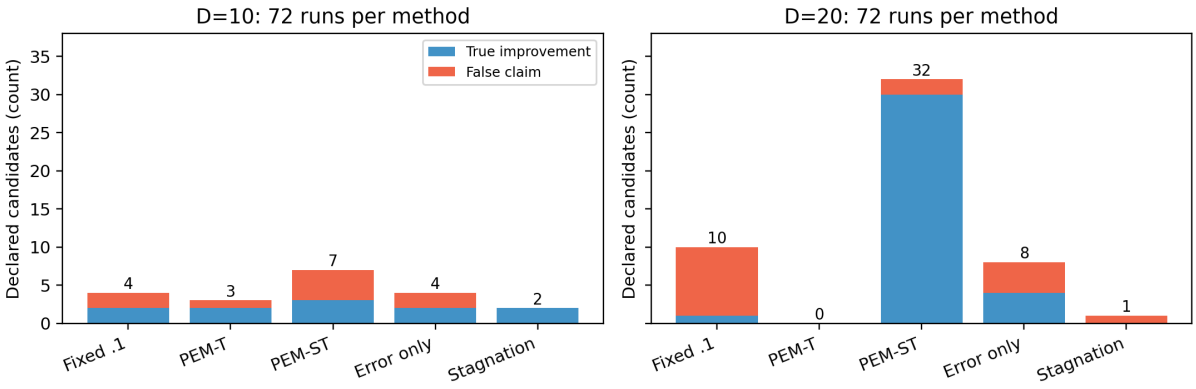


Figure 2: Counts of actual improvement and false claims on declared candidates. A zero column denotes no declarations, not an established reliability advantage.

Failure-only feedback is less detrimental than full feedback on the endpoint summary but still does not achieve an overall benefit over fixed .1. Its combined removal of width and stagnation cannot isolate those terms' independent contributions.

7.4 Common-history diagnostics

Table 4 uses the same T bound within each frozen history/model/pool. Under fixed .3 histories, failures at the actually selected point exceed pool failures. Under greedy histories, their ordering reverses. The observation supports history-dependent differences among monitoring objects and does not support a universal direction of selection-induced undercoverage. Each dimension/history has 288 frozen states; each state contributes one actually selected candidate, 256 pool candidates, and 26 top-10% candidates.

Table 4: Temporal bounds across monitoring objects

D	History	Pool	Top 10%	Actually selected
10	fixed03	0.0425	0.0290	0.1250
10	greedy	0.1070	0.0395	0.0868
10	pool_random	0.0773	0.0944	0.0521
20	fixed03	0.0481	0.0423	0.1424
20	greedy	0.1691	0.0843	0.0938
20	pool_random	0.0935	0.1062	0.1042

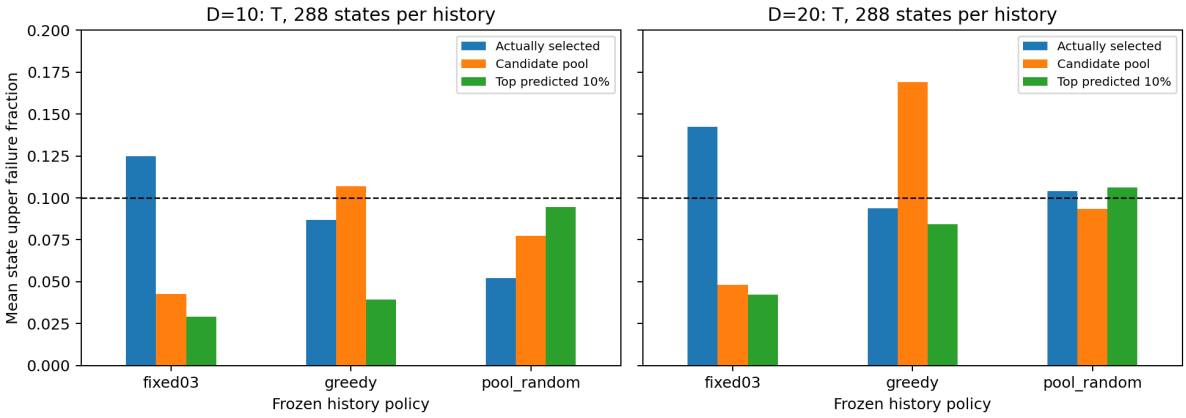


Figure 3: State-averaged one-sided failure fractions under frozen common histories. The dashed target is .1. Pool labels are correlated and are not treated as independent problems.

Table 5 compares calibrators on the actually selected points, with the three histories equally weighted. Spatial localization lowers some diagnostic losses and margins; this is a common-history estimation result, not a closed-loop optimization gain or a coverage guarantee. Approximately one quarter of the dimension-10 checkpoint states fall back because of support, whereas these dimension-20 checkpoints do not.

Table 5: Calibrators on actually selected points

D	Calibrator	Quantile loss	Margin	ESS	Fallback
10	G	0.1355	1.1044	45.5000	0.0000
10	S	0.1241	0.9463	38.8140	0.2500
10	ST	0.1187	0.9175	32.1634	0.2500
10	T	0.1284	1.0262	37.7288	0.2500
20	G	0.1465	0.9365	74.7500	0.0000
20	S	0.1322	0.7874	60.5382	0.0000
20	ST	0.1307	0.7643	42.6863	0.0000
20	T	0.1409	0.9114	52.3172	0.0000

Pool-declaration subsets are also sparse: T has only 13 declared candidates under dimension-10 fixed .3 histories, and one under dimension-20 fixed .3 histories. Full state, nonempty-state, candidate, and

failure denominators are publicly retained in the diagnostic CSV files. We do not infer stable reliability from a small or empty subset.

7.5 Common-state immediate utility

Table 6 reports normalized true improvement at one chosen candidate from the same frozen pool. T upper-bound selection is identical to mean greedy and is not repeated as an independent column. Full feedback has lower mean immediate improvement than mean greedy and than the predeclared fixed .3 rule in all six dimension/history cells.

Table 6: Common-state normalized one-step improvement

D	History	Mean	Fixed .3	Feedback
10	fixed03	0.0260	0.0219	0.0213
10	greedy	0.0310	0.0257	0.0251
10	pool_random	0.1240	0.0925	0.0867
20	fixed03	0.0350	0.0238	0.0172
20	greedy	0.0541	0.0433	0.0381
20	pool_random	0.1326	0.0871	0.0795

Immediate utility does not exclude a delayed benefit from exploration; however, the closed-loop development experiment supplies no evidence of that benefit for this prototype. It is therefore unsupported to state that additional distance exploration repaired the model.

8 Independent mechanism validation

8.1 Protocol and integrity

After the development gate failed, we froze a separate mechanism protocol before generating any labels on its instances. The algorithms, model, candidates, residual weights, support threshold, and feedback coefficients are unchanged. All 24 BBOB functions, dimensions 10/20/40, instances 11–13, and three repeats are included. Six internal configurations yield 3,888 runs and 907,200 paid evaluations. Three preselected history strategies yield 1,944 histories, 7,776 frozen states, and 1,990,656 additional diagnostic labels. Original instances 6–10 remain reserved. Development and validation estimates are never pooled.

The independent audit checks the complete Cartesian design, seeds, unchanged frozen sources, paid prefixes, original signed residuals, training indices that precede each prediction, selected-point/pool agreement, and separate diagnostic archives. There are 0 exact repeated internal calls. Training truncation is actually used in 185328 fitted prediction rounds; this is not evidence of general high-dimensional effectiveness. G/T upper-versus-mean diagnostic index disagreements number 0. The mathematical equivalence below concerns exact real arithmetic, with finite-precision outcomes audited separately.

Three comparisons in each dimension form one nine-test Holm family: the history interaction in selected-minus-pool failures; the PEM-T minus fixed .1 endpoint; and the T-feedback minus mean immediate improvement, with three histories equally weighted. Common-state components are averaged over four states per run, then each component's three-run median is taken before the contrast, followed

by instance and function averages. The 4,000-resample paired hierarchical intervals are not multiplicity-adjusted. Function resampling summarizes heterogeneity over this suite; it does not establish a population model for every black-box task.

Table 7: All nine prespecified comparisons. Endpoint negative favors PEM-T; immediate improvement positive favors feedback. History interaction has no inherent favorable sign.

D	Contrast	Estimate	CI lower	CI upper	Positive	Negative	Holm p
10	History interaction	0.0851	0.0311	0.1234	20	4	0.00149
20	History interaction	0.1225	0.0696	0.1948	21	3	0.00065
40	History interaction	0.1449	0.0941	0.2244	17	7	0.00170
10	Endpoint T minus .1	0.1668	0.0749	0.2391	22	2	5.36e-06
20	Endpoint T minus .1	0.0998	0.0467	0.1905	19	5	0.00170
40	Endpoint T minus .1	0.0639	0.0130	0.1123	19	5	0.00170
10	Immediate feedback minus mean	-0.0170	-0.0431	-0.0067	1	18	0.00078
20	Immediate feedback minus mean	-0.0158	-0.0352	-0.0079	0	21	0.00048
40	Immediate feedback minus mean	-0.0344	-0.0630	-0.0100	1	20	0.00048

8.2 Endpoint and common-state decision effects

The endpoint estimates are D=10: log difference 0.1668, geometric error ratio 1.468, 95% bootstrap interval [0.0749, 0.2391], Holm p=5.36e-06; D=20: log difference 0.0998, geometric error ratio 1.258, 95% bootstrap interval [0.0467, 0.1905], Holm p=0.00170; D=40: log difference 0.0639, geometric error ratio 1.159, 95% bootstrap interval [0.0130, 0.1123], Holm p=0.00170. These ratios summarize equally weighted function-level relative errors and are not ratios shared by every run. Individual-function directions are fully retained in the figure; no function is omitted after observing its result. Immediate feedback-minus-mean effects are D=10: -0.0170 (Holm p=0.00078); D=20: -0.0158 (Holm p=0.00048); D=40: -0.0344 (Holm p=0.00048). These common-state gains concern the next evaluation only and do not identify a long-run causal pathway.

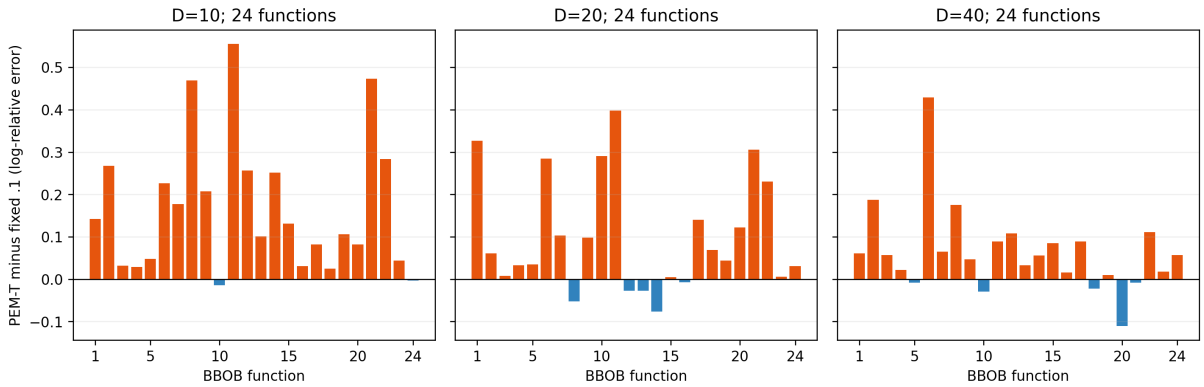


Figure 4: All 24 endpoint contrasts in each dimension. Positive means larger PEM-T errors.

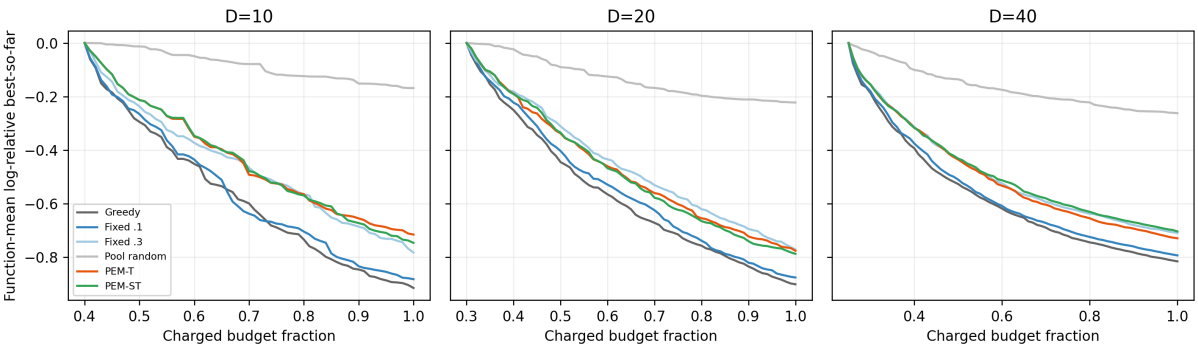


Figure 5: Best-so-far curves after common initialization, with each instance's run median followed by equal instance/function averages. All plotted evaluations are charged.

The frozen secondary AUC floors absolute error at 10^{-12} before division by the initial error; the primary endpoint floors the relative ratio afterward. The curves preserve the former convention and are not numerically forced to match the primary endpoint at near-zero errors. The conventions and affected counts are disclosed in the secondary-metric file. No new significance test is added for AUC or curve inspection.

8.3 History, declarations, and effective support

The history-interaction estimates are D=10: +0.0851 (Holm p=0.00149); D=20: +0.1225 (Holm p=0.00065); D=40: +0.1449 (Holm p=0.00170). This is the fixed .3 selected-minus-pool difference minus the greedy corresponding difference. Across policies, identical seeds do not imply identical candidate pools. These differences describe the coupled history/policy system rather than a fixed-pool causal effect.

Table 8: Temporal pool and actually selected failure fractions, under the prespecified state/run/instance/function aggregation.

D	History	Pool failure	Selected failure
10	fixed03	0.0479	0.1215
10	greedy	0.1018	0.0903
10	pool_random	0.0859	0.0660
20	fixed03	0.0442	0.0833
20	greedy	0.1494	0.0660
20	pool_random	0.0900	0.0729
40	fixed03	0.0505	0.0799
40	greedy	0.1850	0.0694
40	pool_random	0.0978	0.0694

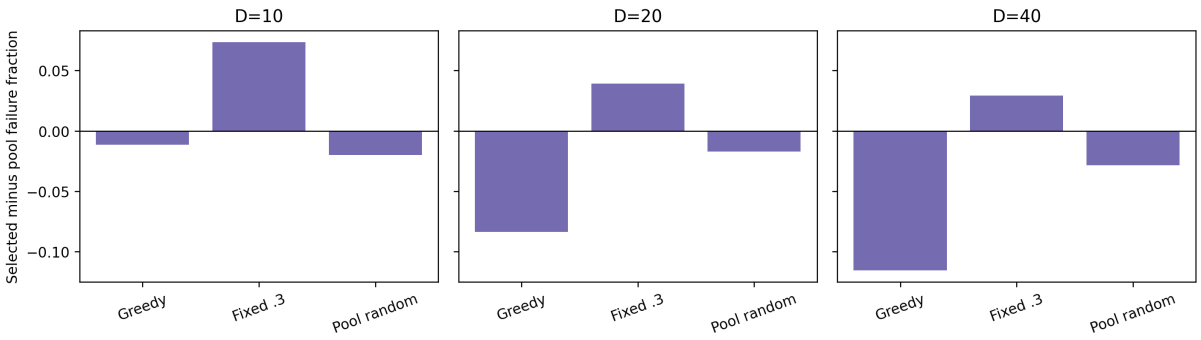


Figure 6: Selected-minus-pool failures by preselected history. A reversal is retained rather than interpreted as universal selection deterioration.

Table 9: Exploration, failures, and declaration denominators in independent closed-loop runs. FDP is pooled false/claims; no-claim FDP is NA.

D	Method	Exploration	Failure	Claims	False	Pooled FDP
10	PEM_ST	0.3936	0.1052	48	23	0.4792
10	PEM_T	0.3967	0.1008	31	20	0.6452
10	fixed01	0.0949	0.0812	35	21	0.6000
20	PEM_ST	0.3687	0.1078	158	36	0.2278
20	PEM_T	0.3644	0.0942	34	27	0.7941
20	fixed01	0.1018	0.0833	48	32	0.6667
40	PEM_ST	0.3701	0.1196	870	98	0.1126
40	PEM_T	0.3435	0.0964	155	30	0.1935
40	fixed01	0.1007	0.0930	388	71	0.1830

For PEM-T, D=10: 31 declarations, 20 false, pooled FDP 64.5%; D=20: 34 declarations, 27 false, pooled FDP 79.4%; D=40: 155 declarations, 30 false, pooled FDP 19.4%. An approximately .1 overall failure fraction does not imply an approximately .1 false-declaration fraction. These small declaration denominators do not establish stable per-declaration probabilities. ST lowers the descriptive selected-point quantile loss in all nine dimension/history cells, yet its closed-loop endpoint remains worse than fixed .1 in every dimension. These are separate diagnostic and trajectory comparisons, not an identified causal relationship.

Table 10: Descriptive state-mean support across the three preselected history policies and four checkpoints.

D	Calibrator	State mean ESS	State mean fallback
10	ST	32.9284	0.2500
10	T	37.7288	0.2500
20	ST	43.6167	0.0000
20	T	52.3172	0.0000
40	ST	49.4855	0.0004
40	T	58.5602	0.0000

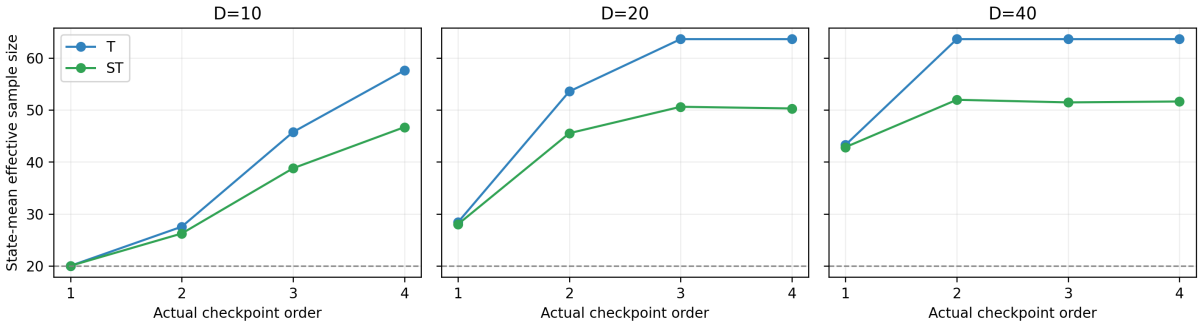


Figure 7: T/ST effective support at the four actual checkpoint orders. The threshold is an engineering guard and does not guarantee coverage.

The first diagnostic checkpoints are 41/100, 61/200, and 101/400. Sparse declared-pool subsets disclose nonempty states, candidate counts, and failures separately. Localized quantile-loss, margin, ESS, and fallback differences and the five fixed BBOB structure groups are available in full secondary tables; no favorable subgroup becomes a new main result.

8.4 External background references

Sobol and pycma CMA-ES retain their own search mechanisms and initialization. Their 1,296 runs and 302,400 paid evaluations passed separate journal audits. For descriptive common-scale comparison only, each external run's final error is divided by the matching internal common-initialization error, and medians are taken after this run-level normalization. The external optimizers never receive those internal labels. This supports budget-matched background comparison, not a same-history causal comparison or reproduction of published SAEA competitors.

Table 11: Descriptive background endpoint scores over all 24 functions. Lower is better; no additional test family is introduced.

D	Fixed .1	PEM-T	PEM-ST	Sobol	CMA-ES
10	-0.8823	-0.7156	-0.7465	-0.0805	-0.4275
20	-0.8761	-0.7763	-0.7877	-0.1066	-0.4799
40	-0.7934	-0.7295	-0.7041	-0.1029	-0.4440

GP-EI and author TuRBO use separate frozen protocols and strict pre-call budget guards. Their complete audited outcomes are now included in Section 9. Author-code restart counterexamples are control-flow checks, not performance observations. Resource precheck instances are excluded from every benchmark table.

9 Complete references, exclusive cost, and application transfer

9.1 Full descriptive BBOB reference evidence

Both frozen GP-EI and author TuRBO-1 batches complete 648 runs and 151,200 paid evaluations each, without reducing functions, repeats, GP training, or acquisition candidates. Together with the six internal and two basic references, all ten methods include 6,480 runs and 1,512,000 paid evaluations. External methods retain their own search and initialization. The matching internal initial-error denominator enters analysis only; normalization precedes within-instance medians. The 24 function scores are equally weighted. These descriptive comparisons add no significance family and do not reproduce the unavailable SAEA competitors.

GP-EI implements classical expected improvement [10] using the explicitly frozen kernel and search; it is not an exact reproduction of original EGO software. TuRBO-1 uses the author local trust-region implementation [11] with an external paid-call guard.

Table 12: Full ten-method function-mean log-relative endpoint scores. Lower is better.

Method	D=10	D=20	D=40
greedy	-0.9152	-0.9014	-0.8158
fixed01	-0.8823	-0.8761	-0.7934
fixed03	-0.7829	-0.771	-0.7097
pool_random	-0.1683	-0.2223	-0.2623
PEM_T	-0.7156	-0.7763	-0.7295
PEM_ST	-0.7465	-0.7877	-0.7041
sobol	-0.0805	-0.1066	-0.1029
cma_es	-0.4275	-0.4799	-0.444
gp_ei	-1.3292	-1.033	-0.9323
turbo_1	-1.102	-1.1219	-1.0389

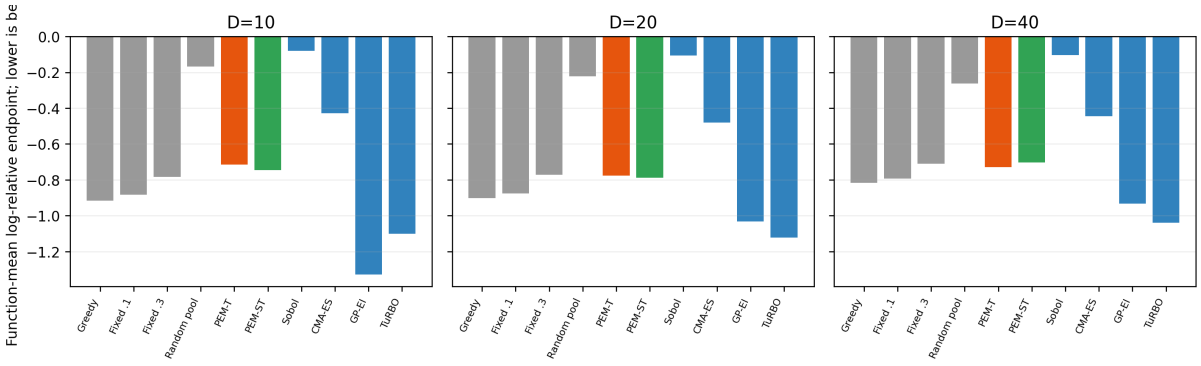


Figure 8: Complete BBOB background reference endpoints; no partial runs or resource-precheck instances.

9.2 Exclusive cost evidence

The frozen resource design contains 54 fresh, serial process measurements on f1/instance4 across D=10/20/40 and three repeats of six methods. Their 12,600 paid calls are reported as cost observations, excluded from benchmark effect estimation. Actual process and background-CPU guards precede measurements. Runtime includes journaling and each method retains its own initialization. Greedy/fixed controls still perform temporal monitoring, so these are implemented-system costs, not stripped-down ideal algorithm costs.

Table 13: D=10; three-run medians; full min/max and per-process observations remain in the supplement CSV.

Method	Algorithm s	Nonobjective s	Peak GiB	Post-init prep s
PEM_ST	0.6547	0.6543	0.0945	0.0108
PEM_T	0.6075	0.6072	0.0947	0.01
fixed01	0.602	0.6016	0.0947	0.0099
gp_ei	27.6737	27.6724	0.2863	0.3456
greedy	0.5983	0.5979	0.0945	0.0098
turbo_1	7.4536	7.4522	0.3905	0.0923

Table 14: D=20; three-run medians; full min/max and per-process observations remain in the supplement CSV.

Method	Algorithm s	Nonobjective s	Peak GiB	Post-init prep s
PEM_ST	1.6909	1.6901	0.0946	0.012
PEM_T	1.5254	1.5246	0.0948	0.0108
fixed01	1.5252	1.5244	0.0948	0.0108
gp_ei	175.864	175.861	0.3059	1.0989
greedy	1.5246	1.5238	0.0945	0.0108
turbo_1	35.0159	35.013	0.5936	0.2183

Table 15: D=40; three-run medians; full min/max and per-process observations remain in the supplement CSV.

Method	Algorithm s	Nonobjective s	Peak GiB	Post-init prep s
PEM_ST	4.7496	4.7476	0.096	0.0157
PEM_T	4.3878	4.3856	0.0962	0.0145
fixed01	4.3787	4.3766	0.0965	0.0145
gp_ei	1308.81	1308.8	0.3073	4.0897
greedy	4.3615	4.3594	0.0958	0.0144
turbo_1	255.912	255.906	1.0791	0.7993

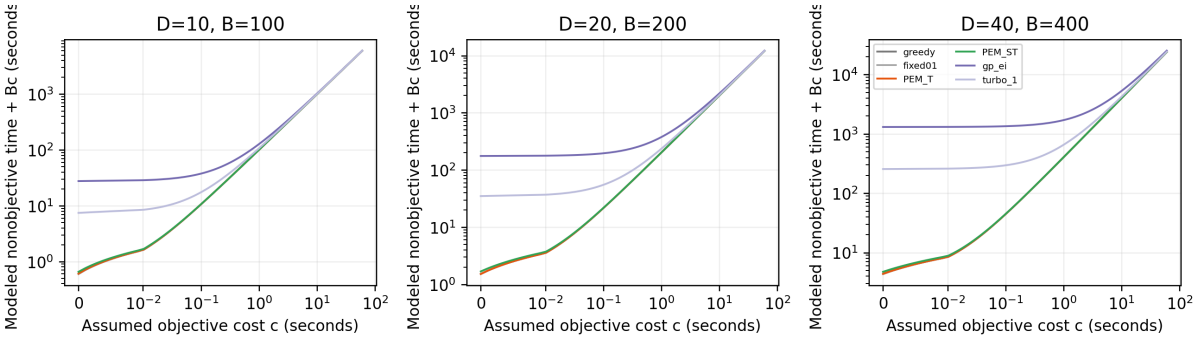


Figure 9: Hypothetical objective-cost scenarios from serial measurements; all methods use the same B in a dimension.

At an assumed objective cost c, total time is modeled as measured nonobjective time plus Bc. This scenario is not an observed application acceleration. Equal evaluation budgets have equal objective-cost slopes and cannot establish evaluation savings.

9.3 LassoBench mechanism transfer

LassoBench [12] provides the application objectives. A separate protocol is frozen before formal seed42 labels. Three original tasks (D=60/100/180), eight methods, twenty optimization repeats, and 500 evaluations yield 480 runs and 240,000 full configuration evaluations (1,200,000 CV-fold solves). Internal initialization is 140/220/380 paid calls; external initialization remains native. The author five-fold full-fidelity objective, fixed data/splits, solver settings, bounds, and target definitions are unchanged.

Unknown optimum is never imputed. The main endpoint is original validation loss. After all 500 labels are saved, the first lowest-validation configuration receives exactly one author test call, adding 480 single fits separately. Synthetic test MSPE uses the author fixed oracle normalizer, while DNA uses absolute test MSE; these scales are not pooled. The synthetic constructor computes its fixed test normalizer, but no test-performance result enters any optimization feedback.

Table 16: synt_simple; twenty optimization-run dispersion and post-completion test medians.

Method	CV median	Q1	Q3	Test MSPE median	F-score median
PEM_ST	0.7896	0.7656	0.8528	1.1435	0.6667
PEM_T	0.7714	0.7557	0.8654	1.2013	0.6667
cma_es	0.758	0.7176	0.7826	1.072	0.5455
fixed01	0.7534	0.7092	0.816	1.1441	0.6667
gp_ei	1.0118	0.9392	1.0847	1.1167	0.6
greedy	0.7501	0.703	0.7755	1.1078	0.6
sobol	1.2057	1.0326	1.2544	1.1865	0.4808
turbo_1	0.6448	0.5888	0.7032	1.0643	0.6

Table 17: synt_medium; twenty optimization-run dispersion and post-completion test medians.

Method	CV median	Q1	Q3	Test MSPE median	F-score median
PEM_ST	1.146	1.1122	1.1731	0.7697	0.7692
PEM_T	1.1519	1.1327	1.1684	0.7562	0.8013
cma_es	1.1888	1.1345	1.237	0.7927	0.625
fixed01	1.1186	1.0751	1.1418	0.7486	0.7692
gp_ei	1.61	1.5226	1.7486	0.8469	0.6667
greedy	1.1103	1.069	1.1616	0.7703	0.7692
sobol	1.8953	1.7099	2.0302	0.8846	0.625
turbo_1	1.0277	0.9575	1.0632	0.835	0.625

Table 18: DNA; twenty optimization-run dispersion and post-completion test medians.

Method	CV median	Q1	Q3	Test MSPE median	F-score median
PEM_ST	0.3071	0.3056	0.3092	0.3304	NA
PEM_T	0.3073	0.3054	0.3093	0.3277	NA
cma_es	0.3014	0.2992	0.3043	0.3285	NA
fixed01	0.306	0.3025	0.3079	0.3307	NA
gp_ei	0.336	0.3328	0.3384	0.3495	NA
greedy	0.3057	0.3042	0.3067	0.327	NA
sobol	0.3348	0.332	0.3375	0.3439	NA
turbo_1	0.3028	0.3005	0.3052	0.3278	NA

Table 19: Prespecified application family, PEM-T minus fixed .1 raw final validation loss. Positive favors the fixed control. Intervals are unadjusted paired-run bootstrap.

Task	Median difference	CI low	CI high	Holm p	Loss	Win	Tie
synt_simple	0.0503	0.0032	0.0725	0.0826	14	6	0
synt_medium	0.0393	0.0221	0.0626	0.0192	16	4	0
DNA	0.0027	0.0005279	0.0044	0.0592	16	4	0

synt_simple: higher, Holm p=0.0826; synt_medium: higher, Holm p=0.0192; DNA: higher, Holm p=0.0592.

The three comparisons form an application-only Holm family, separate from the nine BBOB mechanism tests. Twenty repeats describe optimization randomness on one fixed dataset and split per task, not twenty independent datasets. The Wilcoxon and paired bootstrap assumptions and limits are retained regardless of significance or direction.

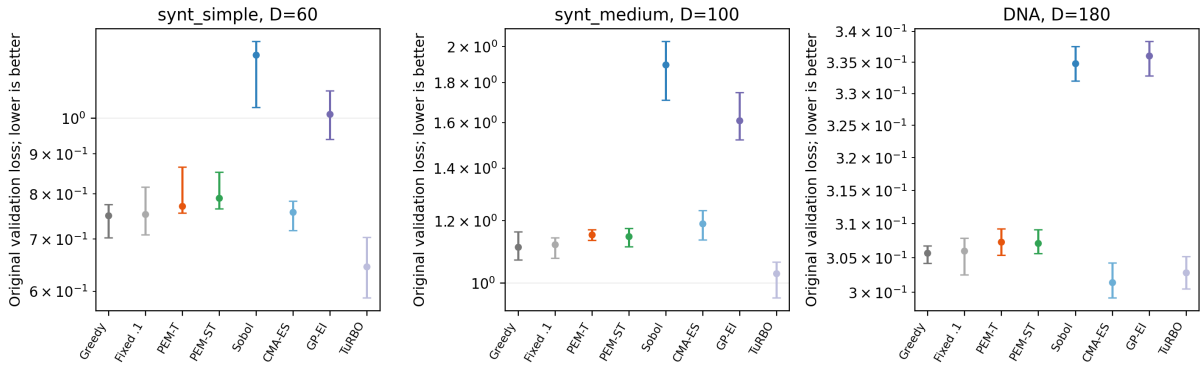


Figure 10: Application original validation endpoints with run-IQR bars; unknown-optimum error is never constructed.

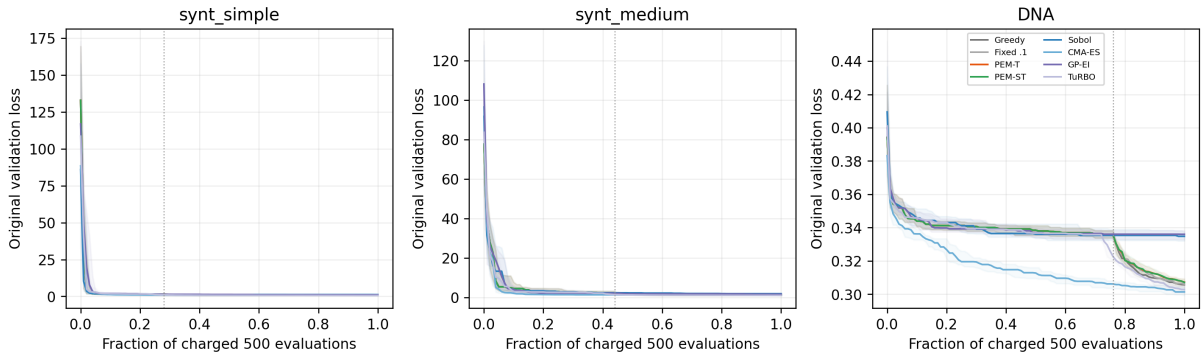


Figure 11: Application original-loss median and IQR curves. The dotted line marks internal initialization; external designs differ. Zero fraction displays the first paid label.

Table 20: Descriptive post-initialization selected-point monitoring; no hidden pool objectives are acquired.

Task	Method	Failure	Q-loss	Claims	False	Pooled FDP	No-claim runs
DNA	PEM_ST	0.0704	0.0698	0	0	NA	20
DNA	PEM_T	0.0629	0.084	0	0	NA	20
DNA	fixed01	0.0346	0.0627	0	0	NA	20
DNA	greedy	0.0079	0.041	1	1	1.0000	19
synt_medium	PEM_ST	0.1036	0.0778	53	46	0.8679	1
synt_medium	PEM_T	0.0887	0.0904	7	6	0.8571	14
synt_medium	fixed01	0.0505	0.0452	27	23	0.8519	5
synt_medium	greedy	0.0134	0.0147	60	54	0.9000	1
synt_simple	PEM_ST	0.1103	0.0946	75	74	0.9867	1
synt_simple	PEM_T	0.0931	0.1104	7	6	0.8571	15
synt_simple	fixed01	0.0549	0.0476	76	74	0.9737	1
synt_simple	greedy	0.0367	0.0137	220	218	0.9909	0

These closed-loop histories differ between methods and do not identify same-pool causal effects. Zero declarations give NA FDP, with counts disclosed. The 300-dimensional synt_high extension therefore uses its separately frozen 1,000-evaluation protocol, reported in Section 11. Application findings cannot undo the failed original development gate.

9.4 Finite-element engineering transfer

EngiBench Beams2D v1 [13] supplies a deterministic finite-element compliance evaluation on a 50 by 100 mesh. Directly treating all 5,000 element densities as optimization variables would be incompatible with the present initialization and surrogate. We therefore define a limited engineering design family: 19 effective logits and one fixed anchor form a 4 by 5 coarse field, cubic interpolation expands it to the finite-element grid, and a unique scalar shift fixes the mean sigmoid density at 0.35. The evidence concerns this 19-dimensional family, not general 5,000-dimensional topology optimization.

Before formal labels, we freeze five methods, ten independent repeats, and 200 evaluations per run. Greedy, fixed01, and PEM-T share the same initialization and candidate framework and form the mechanism comparison. GP-EI and author-code TuRBO-1 retain native initialization and are budget-matched descriptive references. The resulting 50 trajectories contain 10,000 PRE/LABEL pairs, zero failures, and maximum volume-fraction error 1.67×10^{-16}. Fifty post-completion endpoint re-evaluations reproduce the stored values; they are excluded from algorithm budgets. No hypothesis test is added.

Table 21: finite-element endpoint compliance; lower is better, ten optimization repeats per method.

Method	Median	Q1	Q3	Median improvement from native initialization
greedy	1445.1	1300.8	1694.9	0.362
fixed01	1435.6	1354.2	1580.7	0.383
PEM-T	1972.8	1759.3	2168.9	0.251
GP-EI	2117.5	1428.0	2383.3	0.208
TuRBO-1	578.2	565.1	726.9	0.725

For the paired internal comparison, PEM-T minus fixed01 has median compliance difference +488.7 (IQR +134.7 to +814.7), with PEM-T better in 2/10 and worse in 8/10 repeats. Against greedy, the median difference is +575.2 and the direction count is again 2/8. This engineering transfer therefore strengthens the failure boundary rather than rescuing the feedback rule. TuRBO-1 is the strongest descriptive method, but its native initialization prevents interpreting that comparison as a same-history causal effect.

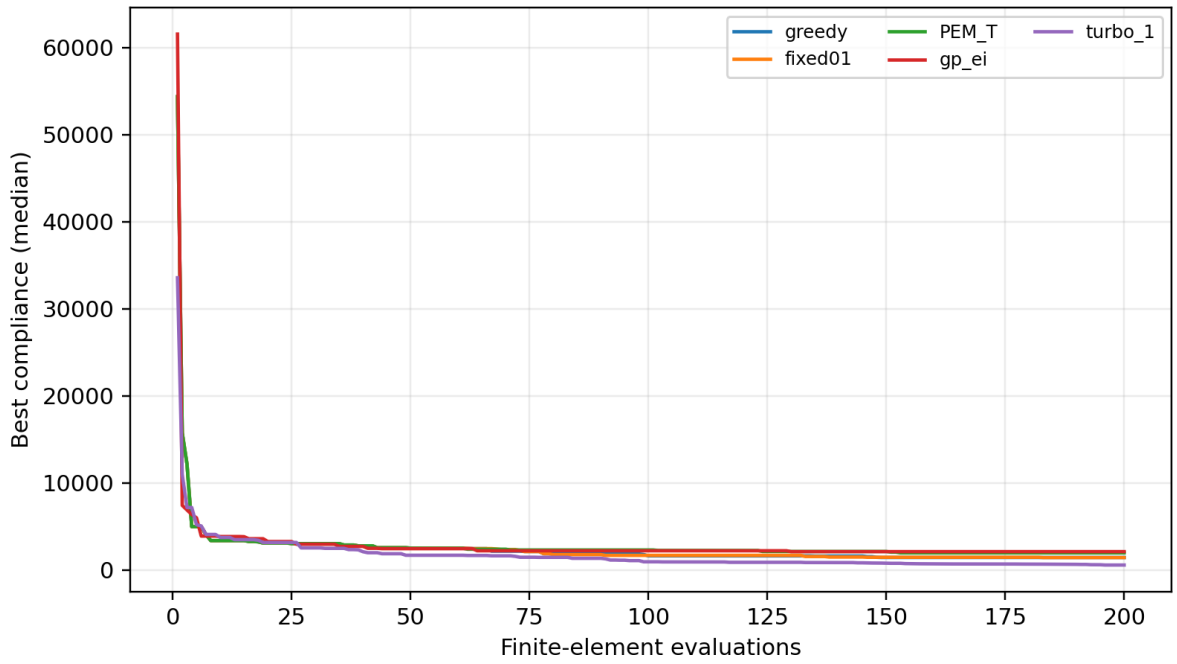


Figure 12: Median best-so-far finite-element compliance over 200 paid evaluations.

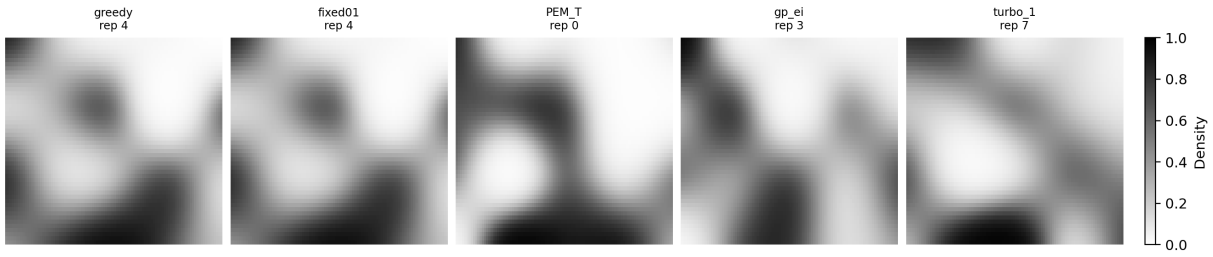


Figure 13: Representative median-run density fields within the frozen 19-dimensional design family.

10 Discussion

The retained evidence has two distinct meanings. Ordering redundancy is a deterministic property of the stated rule, supported by trajectory audits. Increased exploration without the required development benefit is an empirical finding about this feedback design, model, candidate mechanism, and budget. It does not imply that all uncertainty feedback or conformal optimization is ineffective.

Differences between pool and selected-point failures depend on history. Fixed mixing can select either a mean-greedy candidate or a spatially distant candidate. Labeling every such difference as greedy selection bias obscures the exploration policy's role. Local residual weighting may improve a diagnostic loss without establishing that the region contains valuable optimization opportunities or that paying to evaluate it improves future decisions.

The upper-bound margin, failure rate, declaration frequency, and final objective must be read together. A low failure rate can be achieved by wide bounds or nearly absent declarations. In this study, the main prototype approximately attains its target observed failure fraction while making very few declarations and producing worse development endpoints. This separates empirical monitoring quality from useful budget management.

The recorded implementation costs include fitting, candidate generation, bound computation, selection, objective calls, and journaling. Shared-process elapsed times are not exclusive wall-clock comparisons. The analytic benchmark is extremely cheap and cannot establish application acceleration. Greedy/fixed variants' recorded costs include T monitoring that does not affect selection. Dense RBF fitting, training truncation, and candidate generation should be evaluated separately, alongside peak process memory, before practical time claims.

The present residual quantile provides no pointwise 90% success probability, selection-conditional coverage theorem, FDP/FDR guarantee, regret bound, or global convergence theorem. Cross-model scale migration and sparse short-budget feedback remain limitations. Development dimensions 10 and 20 do not establish general high-dimensional capability.

The independent mechanism validation is complete and reported separately from development. The original optimization-improvement gate remains failed, regardless of directions on later instances. GP-EI/TuRBO, exclusive resource, three LassoBench tasks, and the finite-element engineering transfer are reported in Section 9; the predefined high-dimensional extension is reported in Section 11. Reliable implementations of several named competitive SAEAs remain unavailable, which limits direct superiority claims but does not make the mechanism and negative-result evidence incomplete.

The real-arithmetic ordering result does not promise identical float64 tie outcomes for arbitrary magnitudes. For example, adding 10^{16} to means [1,0] collapses their computed difference and changes the first-index minimizer. This numerical teaching example is not a benchmark result. Observed trajectory identities and the independent diagnostic index audit provide empirical implementation evidence for their respective tested scopes.

11 Predefined 300-D Extension

The predefined `synt_high` extension used 1,000 paid evaluations because the unchanged internal initialization requires 620, exceeding the 500-evaluation primary-application budget. Eight methods completed 20 optimization runs on one fixed seed42 data set and split (160 runs; 160,000 validation evaluations; 800,000 CV-fold solves), followed by exactly one held-out test call per completed run. The low-effective-

dimensional synthetic construction and different budget preclude a general high-dimensional or cross-task superiority claim.

11.1 Endpoints and convergence

Method	Validation median [Q1, Q3]	Test MSPE median [Q1, Q3]	Test F-score median [Q1, Q3]
greedy	1.38 [1.334, 1.435]	1.508 [1.466, 1.608]	0.6897 [0.6522, 0.7186]
fixed01	1.392 [1.345, 1.442]	1.615 [1.484, 1.677]	0.6977 [0.6596, 0.7317]
PEM_T	1.364 [1.338, 1.439]	1.58 [1.503, 1.653]	0.7143 [0.678, 0.7548]
PEM_ST	1.372 [1.312, 1.476]	1.614 [1.525, 1.653]	0.6977 [0.6818, 0.7317]
sobol	18.54 [16.12, 22.64]	13.45 [10.07, 19.15]	0.3314 [0.3055, 0.354]
cma_es	1.145 [1.089, 1.179]	1.696 [1.612, 1.909]	0.6522 [0.6, 0.6667]
gp_ei	17.04 [14.89, 21]	13.5 [9.747, 19.81]	0.359 [0.3123, 0.3709]
turbo_1	1.157 [1.128, 1.265]	1.654 [1.6, 1.798]	0.723 [0.6977, 0.75]

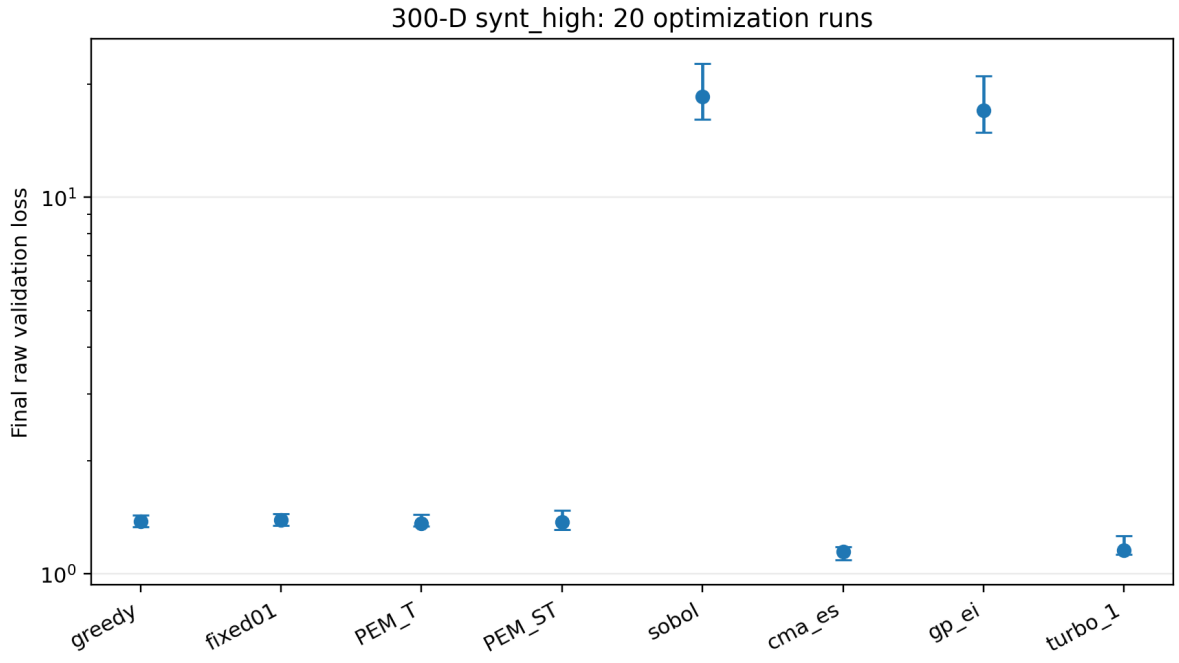


Figure 14: High-dimensional endpoints

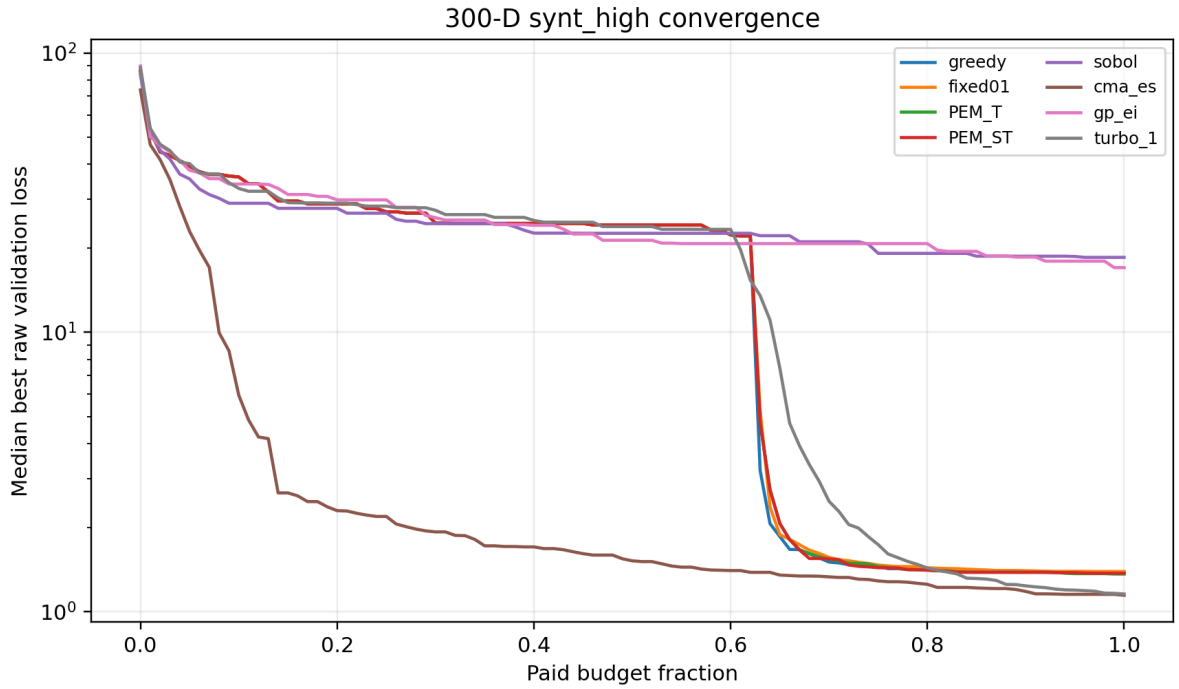


Figure 15: High-dimensional convergence

The prespecified descriptive PEM_T-minus-fixed01 median raw-loss difference was -0.01788, with paired-run bootstrap interval [-0.06477, 0.05017] and 11/9/0 negative/positive/zero differences. No p-value was computed and this contrast was not added to the three-test primary application family. The interval conditions on one fixed data set and split.

11.2 Selected-point feedback and resource boundary

Method	Failure	QLoss	Explore	Fallback	Claims	FDP
greedy	0.07447	0.01695	0	0	257	1
fixed01	0.07895	0.03878	0.09618	0	127	0.9921
PEM_T	0.08921	0.07535	0.3405	0	5	1
PEM_ST	0.1266	0.06124	0.3854	0.01184	119	0.9916

Feedback summaries cover the 380 post-initialization selections per internal run. They describe each closed-loop history and do not establish same-pool causal effects or conformal error control. Before formal labels, eight separate full-budget resource runs completed 8,000 validation evaluations and zero test calls. Their mixed-load timings supported feasibility only; formal optimization times are reported as observed process costs rather than exclusive speedups.

12 Prespecified Model and Training-Cap Sensitivity

Five preselected BBOB structure representatives (f1, f6, f10, f15, f20), dimensions 20/40, instances 14–16, and three optimization repeats evaluated PEM_T under the base setting and four one-factor pertur-

bations. All 450 runs and 135,000 paid evaluations completed; no hidden diagnostic labels or inferential tests were added. This local sensitivity study does not select a replacement default.

D	Variant	Endpoint median [Q1, Q3]	AUC median [Q1, Q3]
20	base	-1.536 [-2.168, -0.8568]	-0.962 [-1.292, -0.5037]
20	cap128	-1.439 [-1.903, -0.7494]	-0.9267 [-1.28, -0.5181]
20	cap512	-1.536 [-2.168, -0.8568]	-0.962 [-1.292, -0.5037]
20	scale125	-1.085 [-1.722, -0.6441]	-0.6593 [-1.087, -0.3838]
20	scale500	-1.473 [-2.095, -0.6943]	-0.776 [-1.079, -0.4353]
40	base	-1.275 [-1.676, -0.8544]	-0.8335 [-1.096, -0.5196]
40	cap128	-0.9176 [-1.179, -0.5716]	-0.6855 [-0.8808, -0.4435]
40	cap512	-1.235 [-1.69, -0.8008]	-0.8185 [-1.123, -0.5374]
40	scale125	-0.8456 [-1.18, -0.5364]	-0.5406 [-0.7806, -0.3901]
40	scale500	-1.151 [-1.311, -0.7564]	-0.7301 [-0.9111, -0.4361]

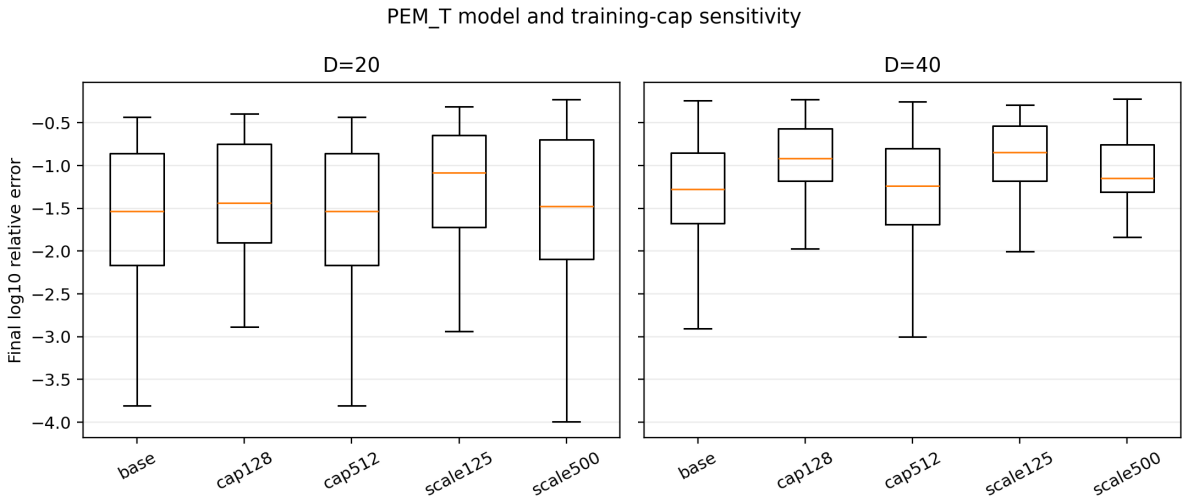


Figure 16: Model and training-cap sensitivity

Paired differences below are variant minus base in final log10 relative error. Negative values favor the variant descriptively.

D	Variant	Difference median [Q1, Q3]	Negative	Positive	Zero
20	cap128	0.06854 [0, 0.1836]	11	30	4
20	cap512	0 [0, 0]	0	0	45
20	scale125	0.2805 [0.08773, 0.5608]	8	37	0
20	scale500	0.09344 [-0.04563, 0.2851]	14	31	0
40	cap128	0.3284 [0.0957, 0.4718]	4	41	0
40	cap512	0.005203 [-0.02706, 0.08161]	18	27	0
40	scale125	0.3253 [0.03915, 0.656]	8	37	0
40	scale500	0.09029 [-0.002699, 0.2816]	12	33	0

The 45 D=20 base/cap512 pairs had identical PRE/LABEL trajectories and candidate arrays, as neither cap was reached. Thus that equality is an implementation invariant rather than evidence that caps never matter. At D=40, cap512 is untruncated only within this 400-evaluation scope.

D	Variant	Capped	Failure	Explore	Fallback
20	base	0	0.08746	0.2789	0
20	cap128	3195	0.1332	0.3444	0
20	cap512	0	0.08746	0.2789	0
20	scale125	0	0.1002	0.3173	0
20	scale500	0	0.07175	0.283	0
40	base	6435	0.08889	0.2541	0
40	cap128	12195	0.1893	0.5487	0
40	cap512	0	0.08993	0.2591	0
40	scale125	6435	0.09733	0.2797	0
40	scale500	6435	0.08674	0.2902	0

Feedback metrics follow each variant's distinct closed-loop history. The limited function/instance set, shared initialization, and absence of multiplicity-adjusted inference restrict these results to robustness description.

13 Prespecified Budget-Performance Sensitivity

13.1 Design and aggregation

Five prespecified BBOB structure representatives (f1, f6, f10, f15, and f20), dimensions 20/40, instances 17–19, and three repeats were run independently at total budgets 5D, 10D, and 15D. Six internal configurations and four external references completed all 2,700 runs and 810,000 paid evaluations, with no diagnostic labels. Each endpoint uses the matching fixed01 common-initialization error as a post-run denominator. We first take the repeat median within each function-instance cell, then average the three instances within each function and the five function scores with equal weight. No hypothesis test, confidence interval, or parameter selection was added.

13.2 Endpoints and convergence

Lower values are better. Each table entry is the prespecified hierarchical mean final log10 relative error.

Method	D20-5D	D20-10D	D20-15D	D40-5D	D40-10D	D40-15D
greedy	-0.961	-1.768	-2.097	-0.9999	-1.544	-1.643
fixed01	-0.9322	-1.749	-2.035	-0.9727	-1.53	-1.664
fixed03	-0.6895	-1.616	-2.041	-0.7998	-1.491	-1.634
pool_random	-0.08979	-0.3369	-0.433	-0.187	-0.3509	-0.4963
PEM_T	-0.7538	-1.689	-1.987	-0.8891	-1.477	-1.598
PEM_ST	-0.7407	-1.676	-2.039	-0.8906	-1.456	-1.57
sobol	-0.1024	-0.1513	-0.209	-0.0429	-0.09157	-0.09792
cma_es	-0.4835	-0.7414	-1.068	-0.4553	-0.6883	-0.949
gp_ei	-1.642	-2.388	-2.574	-1.651	-1.993	-2.174
turbo_1	-1.08	-2.216	-2.502	-0.9778	-1.943	-2.516

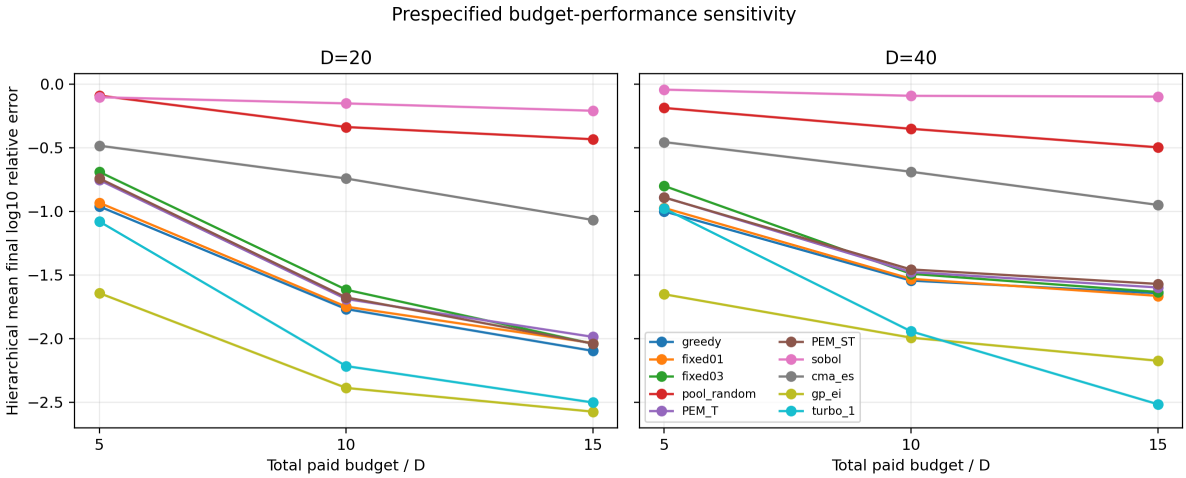


Figure 17: Budget-sensitivity endpoints

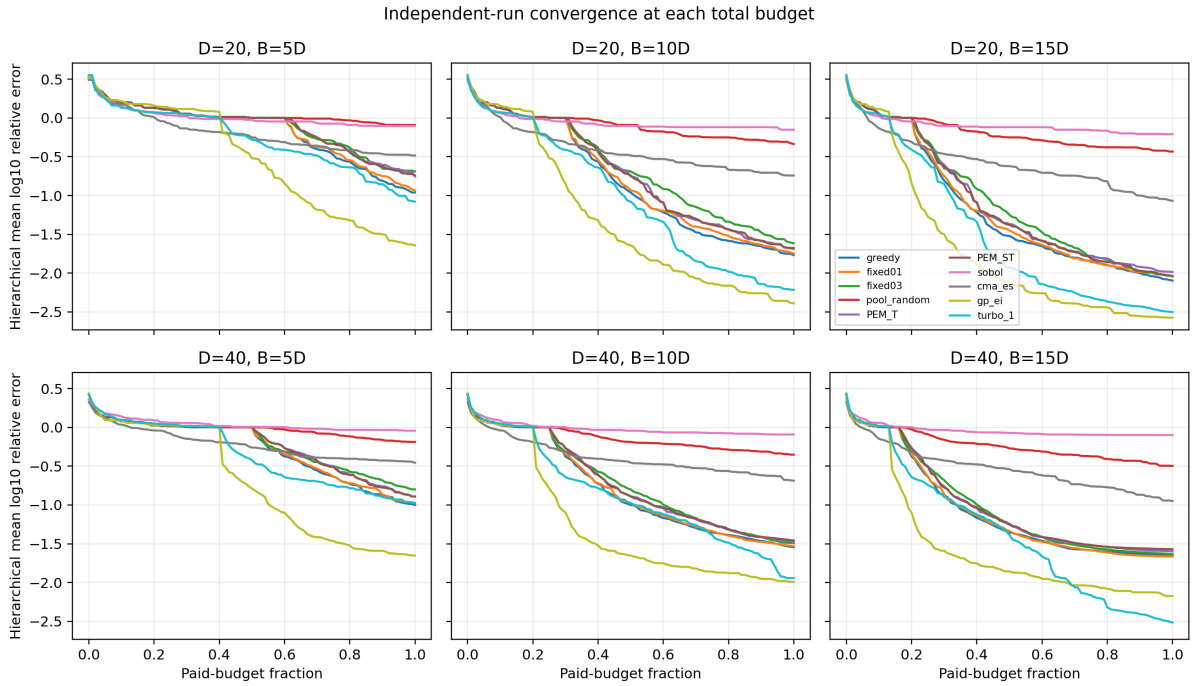


Figure 18: Budget-sensitivity convergence

Each budget is a separate optimization run. A longer-budget endpoint therefore supplies robustness evidence under a larger total allowance; it is not a continuation selected after observing the shorter run. The restricted five-function, two-dimension design and descriptive aggregation do not establish a new general ranking.

13.3 Prefix and information-order audit

Method	Equal 5D/10D/15D prefixes	Required invariant
greedy	90/90	yes
fixed01	90/90	yes
fixed03	90/90	yes
pool_random	90/90	yes
PEM_T	90/90	yes
PEM_ST	90/90	yes
sobol	90/90	no
cma_es	90/90	no
gp_ei	0/90	no
turbo_1	0/90	no

The budget-blind internal core was required to reproduce identical 5D and 10D prefixes inside the corresponding longer run; all 540 required comparisons passed. External prefix equality was recorded rather than required: 180/360 comparisons were equal. That count is an implementation observation, not an assumption used to validate external methods. Full audits replayed all 810,000 PRE/LABEL pairs, internal bounds and management states, and 7,020 raw-file hashes. No total-budget value entered the internal proposal rule.

13.4 Recovery accounting and evidence boundary

The budget-sensitivity interruption left 8 incomplete units with 1,791 physically acquired labels. Their complete directories were archived before resumption, and each affected unit was rerun from its frozen seed. These labels never entered the 2,700 canonical trajectories. The budget-sensitivity physical ledger is therefore 811,791 calls: 810,000 canonical evaluations plus 1,791 disclosed interrupted-run calls. The additional calls are excluded from endpoints, convergence curves, and algorithm budgets.

The completed GP-EI performance batch contains 151,200 canonical algorithm evaluations. During interruption recovery, one GP-EI point (evaluation 349 of `gp_ei_f02_d40_i13_r2_b400`) was physically queried a second time because the newer partial journal contained 348 labels while an earlier archive contained 349. The point and returned label matched exactly; the label entered the optimizer history once, so the canonical 400-step trajectory and endpoint are unchanged. The physical GP-EI ledger is therefore 151,201 calls: 151,200 canonical evaluations plus one disclosed recovery-validation call. The extra call is excluded from algorithm budgets, performance curves, and exclusive-cost analysis. The read-only recovery audit made zero objective calls and verified eight recovered trajectories (2,800 canonical evaluations), 2,014 saved second-generation labels, and 786 second-generation physical calls, one of which overlaps the earlier archive.

This section does not reverse the failed development benefit gate. The paper is positioned as a mechanism and negative-result study and makes no general superiority claim. Full CEC confirmation is not required for that scope. A CEC batch started on 24 September 2026 was stopped after the scope decision; its 30 complete and 8 partial trajectories are retained as exploratory records and excluded from manuscript statistics. Verifiable implementations of the named competitive SAEAs remain unavailable and therefore delimit competitive claims.

14 Conclusion

We implement a prequential one-sided error monitor and examine its decision consequences through isolated diagnostics and closed-loop development runs. Candidate-independent additive corrections preserve the predicted-mean ordering under the specified rule. Diagnostics also show that differences among monitoring objects depend on the history policy. The main feedback prototype does not meet the predeclared development optimization-benefit gate, so we retain a mechanism and failure analysis rather than an optimization-improvement claim. Independent mechanism validation, GP/background references, exclusive costs, three LassoBench tasks, a finite-element beam study, a 300-dimensional extension, and prespecified budget-performance evidence bound the result across settings. The finite-element transfer again places PEM-T behind fixed exploration in 8/10 paired repeats, while TuRBO-1 is the strongest descriptive reference. The contribution is a reproducible mechanism explanation, stopping decision, and negative result, rather than a claim of superior optimization performance.

References

[1] S. Deshpande, C. Marx, V. Kuleshov. Online Calibrated and Conformal Prediction Improves Bayesian Optimization. AISTATS, PMLR 238, 1450–1458, 2024. https://proceedings.mlr.press/v238/deshpande24a.html.

[2] D. Kim, M. Zecchin, S. Park, J. Kang, O. Simeone. Robust Bayesian Optimization via Localized Online Conformal Prediction. arXiv:2411.17387v2, 2025. https://arxiv.org/html/2411.17387v2.

[3] E. Nardone, C. De Stefano, A. Scotto di Freca, F. Fontanella, T. D'Alessandro. Toward Reliable Uncertainty Quantification in Surrogate-Assisted Evolutionary Algorithms via Temporal Conformal Prediction. EvoApplications, LNCS 16524, 384–401, 2026. DOI:10.1007/978-3-032-23604-3_24.

[4] Y. Jin, Z. Ren. Confidence on the Focal: Conformal Prediction with Selection-Conditional Coverage. arXiv:2403.03868v3. https://arxiv.org/html/2403.03868v3.

[5] R. Zhou, C. Ren, Z. Meng, H. Zhu. Surrogate-Assisted Differential Evolution With Search Space Tightening for High-Dimensional Expensive Optimization Problems. IEEE Transactions on Systems, Man, and Cybernetics: Systems 55(10), 7356–7368, 2025. DOI:10.1109/TSMC.2025.3582897.

[6] H. Zou, S. Zeng, C. Li, R. Jiao, F. Zhao. Uncertainty estimation by boosting with Gaussian process for expensive optimization problems. Applied Soft Computing 180, 113320, 2025. DOI:10.1016/j.asoc.2025.113320.

[7] L. Yu, Y. Guo, L. Feng, Z. Zhu, Z. Wei, Y. Hou. A multi-source surrogate-assisted evolutionary algorithm for expensive optimization: An optinformatics perspective. Swarm and Evolutionary Computation 107, 102491, 2026. DOI:10.1016/j.swevo.2026.102491.

[8] COCO. BBOB test suite documentation. https://numbbo.github.io/coco/testsuites/bbob. Accessed 17 September 2026.

[9] IOHprofiler. IOHexperimenter documentation and implementation. https://github.com/IOHprofiler/IOHexperimenter/blob/master/ioh/README.md. Experimental version 0.3.22.

[10] D. R. Jones, M. Schonlau, W. J. Welch. Efficient Global Optimization of Expensive Black-Box Functions. Journal of Global Optimization 13, 455–492, 1998. https://doi.org/10.1023/A:1008306431147.

[11] D. Eriksson, M. Pearce, J. Gardner, R. D. Turner, M. Poloczek. Scalable Global Optimization via Local Bayesian Optimization. Advances in Neural Information Processing Systems 32, 2019. https://proceedings.neurips.cc/paper/2019/hash/6c990b7aca7bc7058f5e98ea909e924b-Abstract.html.

[12] K. Šehić, A. Gramfort, J. Salmon, L. Nardi. LassoBench: A High-Dimensional Hyperparameter Optimization Benchmark Suite for Lasso. Proceedings of the First International Conference on Automated Machine Learning, PMLR 188:2/1–24, 2022. https://proceedings.mlr.press/v188/sehic22a.html.

[13] F. Felten, G. Apaza, G. Bäunlich, C. Diniz, X. Dong, A. Drake, M. Habibi, N. Hoffman, M. Keeler, S. Massoudi, F. VanGessel, M. Fuge. EngiBench: A Framework for Data-Driven Engineering Design Research. Advances in Neural Information Processing Systems, 2025. https://openreview.net/forum?id=YowD33Q89V.

Reproducibility and completion status

Frozen protocols, source fingerprints, dependency locks, seeds, PRE/LABEL journals, candidate pools, separate diagnostic labels, statistics, and failures are archived. Diagnostic labels never enter optimizers. Repeated executions do not enlarge the optimization sample. Budget counterexamples and post-hoc illustrations are identified separately. The predefined high-dimensional extension is reported in Section

11. Reliable competitive SAEA implementations remain unavailable, so no direct superiority claim is made. The failed development gate motivates the mechanism and negative-result scope. The stopped CEC exploration is archived separately and excluded from manuscript statistics. Source, LaTeX, and PDF integration are regenerated for this scope.